

Ashmit Dutta

612-323-1292

✉ ashmitd2@illinois.edu

🌐 ashmitdutta.xyz

📘 [ashmitdutta](#)

🔗 [aedutta](#)

EDUCATION

University of Illinois Urbana-Champaign

BSc, Computer Engineering; Minors in Math & Physics

Urbana, IL

Aug 2023 – May 2027

EXPERIENCE

John Deere

Software Engineer, Part-Time Student

Champaign, IL

Jan 2025 – Jan 2026

- Shipped React + Spring Boot exporting platform **to production** for 500+ internal users on an Agile team.
- Built core REST APIs and CI/CD pipelines over PostgreSQL and DB2, handling millions of records.
- Integrated AWS CloudWatch observability and secured sensitive export data on AWS S3.

Assured and Trustworthy AI Research Lab, UIUC

Machine Learning Research Assistant (Prof. Huan Zhang)

Urbana, IL

Dec 2025 – Present

- Contributing Python and CUDA features to **auto_LiRPA**, the core verification library behind α, β -CROWN (VNN-COMP 2021–2025 winner) used for proving safety of learned controllers and perception models.
- Engineering evaluation pipelines and benchmark datasets for the lab's VNN-COMP 2026 global submission.

University of Chicago Data Science Institute — DSI Summer Lab

Machine Learning Research Intern (Fermilab collaboration)

Chicago, IL

Jun 2024 – Aug 2024

- Selected for DSI's competitive summer research program; built a reusable multi-modal Graph Neural Network library in PyTorch Geometric for Fermilab's Exa.TrkX neutrino-reconstruction project.
- Engineered an end-to-end training pipeline over 20,000+ detector events (H5Py) on a Slurm HPC cluster; presented results at the DSI summer research symposium.

PROJECTS

Stanford TreeHacks: Multi-Modal Video Search Engine | Python (Modal, BrightData, FastAPI)

Feb 2026

- **Winner (Modal Inference Track + Bright Data AI) at Stanford TreeHacks 2026.**
- Built a distributed full-stack video search and dataset generation app using cloud GPU inference.
- Designed a multi-modal embedding pipeline fusing vision, audio, and text for semantic retrieval.
- Integrated Bright Data scraping agents for automated data ingestion; deployed on Modal with MongoDB database.

GPU Accelerated Deep Learning Inference (LeNet-5) | C++, CUDA

Feb 2026 – Apr 2026

- Wrote shape-specialized implicit-GEMM CUDA kernels with fused im2col and constant-memory weights.
- Accelerated Conv2 with WMMA Tensor Cores for 6.6× speedup; 12.5 ms inference at batch 10,000.
- Drove design via Nsight Compute roofline/stall analysis; **placed 5/130 in ECE 408 competition.**

Autonomous Drone Racing | ROS 2, Python, C++, PyTorch

Oct 2025 – Dec 2025

- Built a ROS 2 stack for a Crazyflie 2.1 drone using a min-snap planner and geometric controller.
- Tuned controller gains via Bayesian optimizer; **beat 3 paper benchmarks** with less than 5 cm mean gate error.
- Trained a YOLOv8 gate segmentation model on NeRF-rendered data (98% acc.)
- Analyzed flight Rosbags to debug tracking error; designed voltage-battery feedback system to prevent thrust sag.

LEADERSHIP, PUBLICATIONS & AWARDS

Publication: *Is Human-Written Data Enough? Teaching Reasoning to LLMs Without RL or Distillation* — AI for Math Workshop @ ICML 2025; with NVIDIA researchers. First-hand work on how and where LLM reasoning breaks down.

Co-Founder & Lead Organizer, Online Physics Olympiad (OPhO): largest student-run international physics competition — built a 22k+ Discord community, 3k+ Facebook following, and 3k peak annual competitors since 2020; raised \$10k+ in sponsorships from Citadel, Jane Street, and Wolfram.

Awards & Leadership: 2nd Place, Bradley Mottier Innovation Challenge, UIUC ISE (\$2,000) — AR glasses for ASL translation. Executive Board Member (Treasurer), IEEE Chapter at UIUC.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, SystemVerilog, RISC-V Assembly

AI & Agents: LLM APIs, agentic pipelines, tool use, multi-modal retrieval, embeddings, PyTorch, Modal

Backend & Infra: REST APIs, FastAPI, PostgreSQL, MongoDB, Docker, Git, CI/CD, AWS (S3, CloudWatch), Linux

Systems & Hardware: CUDA, ROS 2, RTL design, FPGAs, STM32, embedded firmware, GDB, QEMU